Lecture 14

1. Bases

Def: Let H be a subspace of V. We say that a set of vectors $\beta = \{b_1, \cdots, b_n\}$ is a basis for H if both of the following are true:

$$\begin{cases} \text{Span}\{b_1, b_2, \cdots, b_n\} = H \\ \beta \text{ is linearly independent} \end{cases}$$

Remark: A subspace H can have many bases

Theorem (Spanning Set Theorem)

Let $S = \{\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_n\} \subset V$ and let $H = \text{Span}(S)$

1. If some vector $\vec{v}_k$ is a linear combination of the other vectors in S, then the set obtained by removing $\vec{v}_k$ still spans S
2. If $H \neq \{0\}$, then some subset

of S is a basis of H

4.5.43

1 part | 1 of 1 point

Let V be a nonzero finite-dimensional vector space, and the vectors listed belong to V. Determine whether the statement below is true or false. Justify the answer.

If there exists a set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ that spans V, then $\dim V \leqslant p$.

Choose the correct answer below.

- A. This statement is true. Apply the Spanning Set Theorem to the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ and produce a basis for V. This basis will not have more than p elements in it, so $\dim V \leqslant p$.
- B. This statement is false. The set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ can be considered a subspace of a finite-dimensional vector space, which can be expanded to find the basis for V. This basis will have at least p elements, so $\dim V \geqslant p$.
- C. This statement is false. Apply the Spanning Set Theorem to the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ and produce a basis for V. This basis will have more than p elements in it, so $\dim V > p$.
- D. This statement is true. The set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ can be considered a subspace of a finite-dimensional vector space, which can be expanded to find the basis for V. This basis will have less than p elements, so $\dim V < p$. If $\dim V < p$ is true then $\dim V \leqslant p$ must also be true.